

AI/ML Internship — Task 2 Report

Feature Engineering, Model Optimization & Performance Comparison

Intern: Zaara Riyaz Khan | Organization: Maincrafts Technology | Date: May 2026

1. Objective

The objective of Task 2 was to build an enhanced House Price Prediction system using the California Housing Dataset. Unlike Task 1 which used a single Linear Regression model, Task 2 involved training multiple machine learning algorithms, applying feature scaling, and comparing model performance using standard evaluation metrics.

2. Dataset

The California Housing Dataset was used, loaded directly from the scikit-learn library. It contains 20,640 records with 8 input features and 1 target variable.

Feature	Description
MedInc	Median income of households
HouseAge	Median age of houses in the block
AveRooms	Average number of rooms per household
AveBedrms	Average number of bedrooms per household
Population	Block population
AveOccup	Average household occupancy
Latitude / Longitude	Geographic location of the block
HousePrice (Target)	Median house value (in \$100,000s)

3. Methodology

- Step 1 — Import Libraries:** pandas, numpy, matplotlib, and scikit-learn were imported.
- Step 2 — Load Dataset:** California Housing Dataset loaded using `fetch_california_housing()`.
- Step 3 — Feature/Target Separation:** Input features (X) and target variable (y = HousePrice) were separated.
- Step 4 — Feature Scaling:** StandardScaler was applied to normalize all features to mean=0 and std=1. This is critical because features like Population and Income exist on very different numeric scales.

Step 5 — Train-Test Split: Data was split 80% training / 20% testing using random_state=42 for reproducibility.

Step 6 — Model Training: Three regression models were trained on the same data for a fair comparison.

Step 7 — Evaluation: Each model was evaluated using RMSE and R² Score on the test set.

Step 8 — Visualization: Actual vs Predicted scatter plot was generated for the best model.

4. Models Trained

- **Linear Regression** — Baseline model. Assumes a straight-line relationship between features and target.
- **Ridge Regression (alpha=1.0)** — Adds L2 regularization to reduce overfitting compared to plain Linear Regression.
- **Decision Tree Regressor (max_depth=5)** — Captures non-linear relationships in the data. Depth limited to 5 to avoid overfitting.

5. Results

The following table shows the performance of all three models on the test set:

Model	RMSE	R ² Score
Linear Regression	0.7456	0.5758
Ridge Regression	0.7456	0.5758
Decision Tree (Best)	0.7242	0.5997

6. Conclusion & Best Model Selection

The Decision Tree Regressor (max_depth=5) was selected as the best-performing model. It achieved the lowest RMSE of 0.7242 and the highest R² score of 0.5997, outperforming both Linear Regression and Ridge Regression on the test set. This indicates that the relationship between housing features and house prices contains non-linear patterns that the Decision Tree captures better than linear models.

Linear Regression and Ridge Regression performed identically (RMSE=0.7456, R²=0.5758), which suggests that regularization alone does not significantly improve performance on this dataset. The R² score of ~0.60 across all models indicates moderate predictive power, which is expected given the complexity of housing price prediction.

7. Potential Improvements

- Use Random Forest or Gradient Boosting (e.g. XGBoost) for better accuracy.
- Apply cross-validation instead of a single train-test split for more reliable evaluation.
- Engineer new features such as rooms-per-person or income-to-age ratio.
- Tune Decision Tree hyperparameters (max_depth, min_samples_split) using GridSearchCV.

— Zaara Riyaz Khan | AI/ML Intern | Maincrafts Technology